

ADITYA SAURAV

B.Tech (Electronics & Communication Engineering), BIT Mesra | 2027
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Career Objective

B.Tech (ECE) student at BIT Mesra with strong foundations in Python, Data Structures, and Machine Learning. Experienced in building scalable backend systems and predictive models. Seeking an AI/ML or Backend Engineering internship to apply data-driven problem solving and develop intelligent, production-ready systems.

Education

Birla Institute of Technology, Mesra	2023 – 2027
B.Tech in Electronics and Communication Engineering	
Cumulative GPA: 7.95 / 10 (till 5th semester)	
Class XII (CBSE) — 80.8%	2023
Class X (CBSE) — 77%	2021

Technical Skills

- **Languages:** Python (Advanced), SQL (MySQL)
- **Frameworks & Tools:** FastAPI, Django, Flask, Git, Docker, BI, Excel, Jupyter Notebook
- **Machine Learning:** Supervised & Unsupervised Learning, Neural Networks (CNN, RNN, ANN), NLP
- **Libraries:** Pandas, NumPy, Scikit-learn, PyTorch, Matplotlib, Seaborn
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming

Projects

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|--|---|
| Smart Grocery E-Commerce Backend | <i>Python, FastAPI, PostgreSQL, Redis</i> |
| <ul style="list-style-type: none">• Designed a high-performance RESTful API using FastAPI to manage a catalog of 1,000+ SKUs with optimized relational schemas.• Developed Inventory Logic handling concurrent transactions via database ACID properties to ensure real-time stock integrity.• Implemented JWT-based Authentication and Role-Based Access Control (RBAC) to secure customer data and admin operations.• Integrated a Personalized Recommendation Module using Scikit-learn, applying content-based filtering for user-specific suggestions. | |
| Credit Card Fraud Detection | <i>Python, Scikit-learn, Pandas</i> |
| <ul style="list-style-type: none">• Built an end-to-end ML pipeline to detect fraudulent transactions in highly imbalanced datasets.• Applied SMOTE for oversampling and evaluated models using Precision-Recall curves, achieving a high F1-score.• Optimized classification thresholds to minimize false negatives in a production-simulated environment. | |
| SMS Spam Classifier | <i>Python, NLP, NLTK</i> |
| <ul style="list-style-type: none">• Developed an NLP-based text classifier using TF-IDF vectorization and Multinomial Naive Bayes.• Processed 5,000+ messages through a custom pipeline including tokenization, lemmatization, and stop-word removal.• Achieved 98%+ accuracy in binary classification of spam vs. ham communications. | |
| Dog vs Cat Image Classifier | <i>PyTorch, CNN, Computer Vision</i> |
| <ul style="list-style-type: none">• Engineered a Convolutional Neural Network (CNN) for image recognition, achieving a significant reduction in classification error.• Leveraged Data Augmentation and Dropout layers to mitigate overfitting and improve model generalization on unseen data.• Implemented Transfer Learning using pre-trained ResNet-50, fine-tuning final layers to accelerate convergence.• Utilized TensorBoard for monitoring loss curves and accuracy metrics to ensure optimal hyperparameter tuning.• Optimized performance using batch size and learning rate scheduling to reach peak validation accuracy. | |

Coding & Problem Solving

- Solved **250+ DSA problems** on LeetCode and TakeUForward (Striver's A2Z Sheet).
- Active participant in competitive programming, focusing on Algorithmic Efficiency and Python optimization.

Certifications & Activities

- Completed Python, SQL, and AI Fundamentals (Coursera, HackerRank).
- **Captain of Cricket Team** — Led college team to multiple inter-university tournament victories.
- Participated in regional hackathons and technical symposiums at BIT Mesra.

Interests

Cricket, Team Sports, Wildlife Photography